FACE Prep CAMPUS

# Course 2: Excel Basics for Data Analysis

## Module 1: Introduction to Using Excel Spreadsheets

### Key Concepts & Vocabulary

- **Workbook vs. Worksheet:** A .xlsx file is the Workbook; individual tabs are Worksheets.
- **Cell Reference:** The intersection of a column (letter) and a row (number), e.g., C25.
- **The Ribbon:** The command bar at the top (Home, Insert, Page Layout, etc.).
- **Formula Bar:** The window above the grid used to view or edit the underlying content of a cell.
- **Name Box:** Displays the address of the active cell or a "Named Range."
- **Fill Handle:** The small green square in the bottom-right of a cell used for copying or "Flash Fill."

### Technical Deep-Dive

#### 1. Data Types & Alignment (Crucial for Quizzes)

- **Text:** Aligns **Left**. Used for names, addresses, and IDs.
- **Numbers:** Aligns **Right**. Used for calculations.
- **Dates:** Aligns **Right**. Excel stores dates as sequential serial numbers (e.g., Jan 1, 1900 = 1).
- **Booleans:** Aligns **Center**. (TRUE/FALSE).

#### 2. Essential Operations

- **Flash Fill (Ctrl + E):** AI-driven pattern recognition. Use it to split "First Last" into two columns without formulas.
- **Paste Special (Alt + E + S): * Values:** Removes formulas, keeps results.
  - **Transpose:** Swaps rows for columns.
- **Freezing Panes:** Found in the **View** tab. Essential for keeping headers visible during vertical/horizontal scrolls.

#### 3. Relative vs. Absolute Referencing (The #1 Quiz Question)

- **Relative (A1):** Adjusts when copied (dragged). A1 becomes A2.
- **Absolute ($A$1):** Locks the reference. It never changes when dragged.
- **Mixed ($A1 or A$1):** Locks only the column or only the row.

# Course 2: Excel Basics for Data Analysis

## Module 2: Advanced Formulas and Logical Functions

### Key Concepts & Vocabulary

- **Syntax:** The specific format required for a function (e.g., =FUNCTION(arg1, arg2)).
- **Arguments:** The inputs provided to a function.
- **Nested Function:** A function placed inside another (e.g., IF(IF())).
- **Lookup Value:** The "key" you are searching for in a table.
- **Table Array:** The range containing the data you want to search.

### Technical Deep-Dive

#### 1. Logical Operators

- **IF:** =IF(condition, value_if_true, value_if_false).
- **IFS:** Allows up to **127** different conditions (replaces nested IFs).
- **AND/OR:** Used to test multiple conditions simultaneously.
  - AND requires **all** to be true.
  - OR requires **any** one to be true.

#### 2. Lookup Functions (The Professional Standard)

- **VLOOKUP:** =VLOOKUP(lookup_value, table_array, col_index, [range_lookup]).
  - **Range Lookup:** Use 0 or FALSE for exact matches (IDs). Use 1 or TRUE for ranges (Tax brackets).
  - **Limitation:** Only looks **right**. Breaks if you insert/delete columns.
- **INDEX-MATCH:**
  - MATCH finds the row number.
  - INDEX retrieves the value at that row number.
  - **Benefit:** Can look **left**; faster on datasets $> 100,000$ rows.
- **XLOOKUP:** =XLOOKUP(lookup_value, lookup_array, return_array).
  - Defaults to exact match. No column index numbers needed.

#### 3. Error Handling (Exam Identification)

- **#N/A:** "Not Available." VLOOKUP didn't find the value.
- **#VALUE!:** Wrong type of argument (e.g., trying to add a number to "Hello").
- **#REF!:** A cell referenced in the formula was deleted.
- **#NAME?:** Typo in the function name (e.g., =SUMM()).
- **#DIV/0!:** Division by zero or an empty cell.
- **IFERROR:** Wrap formulas to hide errors: =IFERROR(VLOOKUP(...), "Not Found").

# Course 2: Excel Basics for Data Analysis

## Module 3: Data Analysis using Pivot Tables

### Key Concepts & Vocabulary

- **Pivot Table:** A tool used to summarize, analyze, explore, and present data without changing the source.
- **Aggregation:** The method of summarizing data (Sum, Count, Average, Min, Max).
- **Source Data:** The raw table. It **must** have headers and no empty rows/columns.
- **Slicer:** A visual filter button for Pivot Tables.
- **Pivot Chart:** A visual representation of a Pivot Table that updates dynamically.

### Technical Deep-Dive

#### 1. Pivot Table Fields

1. **Filters:** Limits the entire dataset (e.g., Show only "2025" data).
2. **Columns:** Horizontal groupings.
3. **Rows:** Vertical groupings (The primary axis of analysis).
4. **Values:** The quantitative field (Usually the "Sales" or "Quantity" column).

#### 2. Calculations in Pivot Tables

- **Value Field Settings:** Change "Sum" to "Count" or "Average."
- **Show Values As:** Allows you to show data as **% of Column Total** or **% of Grand Total**—essential for market share analysis.
- **Calculated Fields:** Allows you to create new metrics inside the Pivot Table (e.g., Profit = Revenue - Cost).

#### 3. Refreshing Data

- Pivot Tables **do not update automatically** when source data changes. You must right-click and select **Refresh**.

# Course 2: Excel Basics for Data Analysis

## Module 4: Basic Charts and Data Visualization

### Key Concepts & Vocabulary

- **Data-Ink Ratio:** Edward Tufte's principle: Maximize the ink used to show data, minimize the "junk" (gridlines, 3D effects).
- **Data Series:** A row or column of numbers plotted in a chart.
- **Axes:** The X-axis (Horizontal/Category) and Y-axis (Vertical/Value).
- **Trendline:** A line added to a chart to show the general direction of data.

### Technical Deep-Dive (Quiz Selection Logic)

| Visualization | Use Case |
|---|---|
| **Column/Bar Chart** | Comparison of categories (e.g., Revenue by Region). |
| **Line Chart** | Trends over **Time** (e.g., Monthly Users). |
| **Pie Chart** | Parts of a whole. Use only for $<$ 5 categories. |
| **Scatter Plot** | **Correlation** between two variables (e.g., Ad Spend vs. Revenue). |
| **Area Chart** | Like a line chart but shows the "Volume" under the line. |

#### Chart Elements for Customization

- **Legend:** Explains what the colors/patterns represent.
- **Data Labels:** Shows the exact number on top of a bar or point.
- **Dynamic Titles:** You can link a chart title to a cell using the = sign in the formula bar.

# Course 2: Excel Basics for Data Analysis

## Critical Exam Prep Summary Checklist

### Quantitative Benchmarks for Excel Analysts

1. **Row Limit:** Excel supports **1,048,576** rows. If your data is larger, you must use Power Pivot or SQL.
2. **Precision:** Excel stores numbers with **15 digits** of precision.
3. **Performance:** VLOOKUP is an $O(N)$ operation. In massive datasets, INDEX-MATCH or XLOOKUP can improve calculation speed by **15-20%**.

### The "Logic Map" for Quizzes

- **Need to find a value?** $\rightarrow$ VLOOKUP or XLOOKUP.
- **Need to group data by month?** $\rightarrow$ Pivot Table.
- **Need to handle a #N/A error?** $\rightarrow$ IFERROR.
- **Need to lock a cell?** $\rightarrow$ F4 key (Adds $ signs).
- **Need to show a trend?** $\rightarrow$ Line Chart.
- **Need to count text cells?** $\rightarrow$ COUNTA.